



Vehicle detection and traffic density estimation using ensemble of deep learning models

Usha Mittal¹ · Priyanka Chawla¹ 

Received: 2 March 2021 / Revised: 6 April 2022 / Accepted: 2 August 2022 /
Published online: 27 August 2022

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract

Traffic density estimation can be used for controlling traffic light signals to provide effective traffic management. It can be done in two steps: vehicle recognition and counting. Deep learning (DL) technologies are being explored more and more as CNN grows in popularity. In this study, initially, data was collected from various open-source libraries that is FLIR, KITTI, and MB7500. Vehicles in the images are labelled in six different classes. To deal with an imbalanced dataset, data augmentation techniques were applied. Then, a model based on an ensemble of the faster region-based convolutional neural networks (Faster R-CNN) and Single-shot detector (SSD) were trained on finally processed datasets. The results of the proposed model were compared with base estimators of the FLIR dataset (Thermal and RGB images separately), MB7500, and KITTI dataset. Experimental results depict that the highest mAP obtained was 94% by the proposed Ensemble on FLIR thermal dataset which was 34% better than SSD and 6% from the Faster R-CNN model. Overall, the proposed ensemble achieves better and more promising results as compared to base estimators. Experimental results also show that detection with thermal images was better than visible images. In addition, three algorithms were compared for estimated density and the proposed model shows significant potential for traffic density estimation.

Keywords Detection · Traffic density · Ensemble · SSD · Faster R-CNN · Convolutional neural network · Deep learning

1 Introduction

As the traffic volume increases on roads, traffic management becomes a serious issue for society. Thus, vehicle detection and classification are a demanding research area as vehicles possess various properties ranging from corners, colours, edges, shadow, texture, etc. Detection [18, 34] is the

✉ Priyanka Chawla
priyankachawla.cse@gmail.com

¹ School of Computer Science and Engineering, Lovely Professional University, Phagwara, Punjab, India

combination of two main processes: classification and localization. In classification [23], the class of vehicles present in the image/video is predicted while localization [56] determines the position of vehicles available and presents detected vehicles by drawing a bounding box around them. With the advancement in technology, high-resolution cameras, and high-speed hardware devices can be used in such systems. Green signals can be optimized at traffic intersections by determining the traffic density. It helps in reducing the traffic jams as well as overall waiting time of vehicles can be reduced.

Vision-based vehicle detection algorithms are classified into three categories that are based on motion, handcrafted features and CNN approaches. Techniques based upon *Motion* comprise optical flow, background subtraction as well as frame subtraction, and these techniques can classify and detect moving vehicles only by capturing images/videos from still cameras. *Hand-crafted feature-based approaches* contain Histogram of Oriented Gradients (HOG) [20], SIFT [39], and Harr-like [51]. These methods have low feature representation. Although these methods work well even if the data size is small and doesn't require any special hardware, in machine learning, features need to be extracted by some expert. The efficiency of the algorithm largely depends upon the accuracy of features identification and extraction methods. Deep Learning (DL) models require a huge amount of training data as it learns high-level features from them. It reduces the task of developing feature extraction algorithms. But these algorithms highly depend upon high-end machines as these algorithms perform many matrix multiplication operations. Graphical Processing Units (GPUs) can very efficiently optimize these operations.

The major concern of vehicle detection is lighting conditions as well as the quality of the video feed [30]. Although vehicles can be easily detected during daytime by identifying front, rear, edges, and corners, it is very difficult to extract features during the night. So, in this paper thermal images dataset has been employed to analyse the detection capability and compared with visible image datasets. From the literature, it had been observed that algorithms failed to detect some of the vehicles and assigned wrong labelling. To deal with this issue, a hybrid model was designed to detect vehicles at different aspect ratios and sizes. As well, a majority voting scheme was applied to assign the final label to the vehicle.

Algorithms designed using ensemble learning consist of multiple models to get better performance than could be provided by the single constituted model. Ensemble learning is the process of combining several models like classifiers or experts to solve any computational problem. The aim is to provide a DL model which consists of many learner models and form a strong learner model thus, improving the accuracy. An ensemble based deep learning model, such as Faster R-CNN and SSD, was created and implemented in this study utilizing a stacking method. The following are the key contributions of the proposed work:

- a) Data Augmentation Methods are employed to increase the size of the datasets.
- b) Duplicate object detection problem from Faster R-CNN and SSD is resolved.
- c) Investigation of majority voting classifier based upon an ensemble of DL methods for detection of vehicles is designed and implemented with better accuracy
- d) Computing the number of vehicles and estimating traffic density on road.
- e) Comparison of the proposed model with its base estimators on different benchmarking datasets - FLIR dataset (Thermal and RGB), MB7500, and KITTI.

The rest of the paper is organized into sections as follows: Section 2 gives a detailed literature review. Section 3 illustrates the proposed technique. Experimental results are given in Section 4. Various comparisons and discussions on the results have been explained in Section 5. Finally, section 6 concludes the paper.

2 Related work

Ozkurt et al. [47] presented a model which consisted of three sub-models. Moving Object Detector (MOD) was used for object detection using the background estimation method. Vehicle Identifier was utilized to identify vehicles in a small, middle, and big category using a neural network. Finally, traffic density was estimated using the recognized vehicles in successive frames. The author performed experimental results on the Istanbul Traffic Management Company (ISBAL) dataset and claimed that results were promising. Suryanto et al. [54] proposed a spatial color histogram model in which color distribution and spatial information were encoded. The proposed method used a voting mechanism to locate new objects in the frame and then a back-projection method was utilized to extract the located object from the background.

Kembhavi et al. [32] described a vehicle detector in which a Color Probability Map was used to capture vehicles' and their surroundings' color statistics along with HOG features. Partial Least Square was employed to reduce the dimensionality of features. Arrospeide et al. [7] proposed a log-Gabor descriptor for vehicle verification and claimed that it was better than the Gabor filter. Zhang et al. [60] designed a cascade ensemble classifier with rejection to not decide the case of ambiguity. The dual majority voting method was used to assign the final target. John et al. [31] utilized adaptive fuzzy C-means clustering and convolutional neural network to segment pedestrians from infrared images. Rezaee et al. [52] designed a particle filter-based multi-view tracking system in which color, edge, texture, and motion were utilized to get better accuracy. For occlusion detection and handling, homography relation, inter-frame affine transformation, and robust statistics were used.

With the aid of frame difference, the author [13] presented a framework for recognizing a moving automobile. The binary frontal view was obtained using an asymmetrical filter on the front image of the automobile, and the model of the car was detected using a three-layer limited Boltzmann machine using deep learning. Author [58] worked on a vision-based system for identifying preceding cars on a highway, taking into account diverse circumstances such as low illumination and weather conditions. Al-Zu'bi et al. [5] suggested a 3D image segmentation model in which HMM method in the distributed environment was utilized. Also, the proposed model made use of AES encryption to secure data. The author's experimental results showed that the given model yielded an accuracy of 98.9%.

In a paper [26, 55], the author developed a method for tracking single objects over short periods using thermal pictures. Author [45] also worked on the detection and classification of things on the sea surface using thermal photos obtained by Unmanned Ariel Systems (UAS). It helped in the search for and recovery of marine artifacts. Over a testing dataset, the experimental results revealed 92.5% accuracy. During the day and at night, the author [61] proposed a surveillance system for identifying and classifying cars. Textures, entropy, homogeneity, energy, and contrast were among the criteria employed in the feature extraction. Author [44] suggested a method for vehicle detection and classification based on visible light images and thermal images. In this study, four features homogeneity, contrast, energy, and entropy were extracted. The author's results showed that visible camera images had an accuracy of 92.7% while thermal images had 65.8% accuracy.

Zhang et al. [53] used a deep neural network to work on vehicle detection and categorization. The goal was to find high-level traits from lower-level ones. The deep neural network performed better for car classification with a 3.34% error rate compared to a regular neural network with a 6.67% error rate in the experiments. For vehicle detection, the author [25] suggested an improved Gaussian mixture model with background removal. After that, feature extraction was done with AlexNet and SIFTS (scale-invariant feature transform),

dimensionality reduction was done with PCA and LDA, and finally, classification was done with SVM (support vector machine). The testing findings showed that at FC6 and FC7, improved GMM with AlexNet DNN was more accurate at identifying and classifying cars. Author [27] developed a deep neural network that employed the YOLO technique for detection and the AlexNet DNN for classification. Vehicle classification on dark pictures using scene modification, late fusion approach, and color transformation method was utilized to enhance the capabilities of DNN. Author [29] proposed an ensemble of deep learning methods to detect and classify moving vehicles. In the given model, initially, image processing techniques were applied to improve the quality of input images then steerable pyramid transform (SPT) and weber local descriptor (WLD) were utilized for feature extraction. The proposed model was simulated on two datasets (MIO-TCO and BIT vehicle dataset) of RGB images and achieved an accuracy of 99.28% and 99.13% accuracy.

S et al. [30] proposed a LittleYOLO-SPP based model in which features were extracted with the varying scale of pooling layers. Mean Square Error (MSE) and Generalized IoU (GIoU) functions were used for bounding box regression to improve the detections of the network. The author claimed that the proposed model detected the vehicles in real-time with high accuracy. Zhu et al. [64] proposed multi-sensor multi-level enhanced CNN with consideration of hybrid realistic scene of scales, occlusion, and illumination to improve vehicle detection abilities. The proposed model consisted of a LiDAR Image composite module and enhanced inference head. Experimental results showed that the proposed model achieved reliable and accurate detection results. AlZu'bi et al. [1] performed an extensive literature survey on autonomous vehicles related technologies in which ITS techniques are used. The review integrated and synthesized monitoring AVs using data fusion from different data sources.

Kesarwani et al. [33] suggested a vector-based regression model detect text to generate a quadrilateral bounding box by using centroid-centric vector regression. The study used a hybrid approach of direct and indirect regression in which single-stage detection with standard non-maximal suppression (NMS) and region rejection multi-scale testing was used. Chowdhury et al. [14] proposed a mixture of color spaces and gradient information to detect license plate detection.

Aqel et al. [6] suggested machine learning models for detecting and classifying healthy and diseased leaves. In the proposed model, the k-means clustering model was utilized for segmentation. Author [2] proposed a method for transferring the trained matrices of HMM from one image to another. The author claimed that the proposed model had less processing time and better detection accuracy.

AlZu'bi et al. [3] employed the internet of Multimedia sensors in smart farming to optimize the irrigation process. Image processing concepts along with IoT sensors and ML models were used to make irrigation decisions. Author [4] proposed a 3D-FCM algorithm to extract ROI in various geometrical orientations for an object in 3D volumes.

Author [35] used a catalogue-based strategy to detect autos in UAV (unmanned aerial vehicle) photos. The Author had discussed the present system, which relied on screening activities in which asphalted regions were identified as having the ability to increase the speed and reliability of car detection. Then, using filtering procedures to locate the exact spots of the car, HOG characteristics were recovered, and the 36-directions for a higher similarity value were found. A single car might be seen many times due to the great resolution of the UAV photograph. Points that belonged to the same automobile were combined to minimize redundancy. Finally, the data were classified using SVM. Author [16] was working on a cloud-based intelligent urban video surveillance system for automatic vehicle detection and tracking.

Author [43] developed a HOG-based approach that took into account congested traffic locations. The hybrid DNN was used by Author [38] to detect and classify objects. For feature extraction and compression, the non-negative matrix factorization (NMF) was applied. For abrupt illumination changes and vibration from the camera, the author [10] employed a shadow removal approach and a background Gaussian Mixture Model to identify, track, and classify vehicles into four groups. The Kalman filter was employed for tracking. Author [22] developed a deep Convolutional activation feature for vehicle identification and classification (DeCAF). Visual features were retrieved, and the accuracy of several approaches such as large-scale sparse learning and deep CNN was compared.

The author [62] suggested a model for classifying vehicles based upon their structure. To extract the boundaries, manual segmentation was performed and then features were extracted from them. V.V et al. [19] gave a method to detect vehicles using Fast R-CNN. In this work, images are captured using a camera, and the dataset is balanced with three classes. The model shows acceptable accuracy if images were clear and did not contain any noise. In paper [12], the author has taken sensor-based data for traffic analysis by deploying sensors alongside the road. Then cascade filtering is used to select data from multiple sources. The CNN-based model was used to classify and detect vehicles. This method is more flexible to noise and provides an accuracy of 98%. Author [59] proposed a feature-fused SSD with tracking guided detection optimization (TDO) strategy in which NMS was used at the post-processing. In [48], the Haar feature extraction method was used. Bat algorithm was utilized to select features. Then vehicle detection and classification were done with enhanced CNN and SVM. Author [9] had implemented SSD and MobileNet SSD for vehicle detection and found that SSD has better performance as compared to MobileNet. Although inference time of MobileNet is less than SSD. Author [8] constructed ShuffleDet in which channel shuffling and grouped convolutions to improve the speed performance of the model. In [37], global networks were ensembled with K local expert networks to improve the performance of vehicle classification. The implemented model achieved an accuracy of 97.92%. To resolve the problem of the imbalance dataset, author [42] utilized a semi-supervised model in which DNN was integrated with Generative Adversarial Nets (GAN) for data augmentation. A summary of the literature survey on vehicle detection and classification is given in Table 1.

2.1 Deep learning object detectors

An innovative two-stage object detector, Region-based Convolution Neural Network (R-CNN) was proposed by Girshick in 2014 [34]. It had improved the performance of detection and gave 53.7% mAP on Pascal VOC2010 [50]. He used the concept of transfer learning and ImageNet architecture was used as the backbone. Prompted by the estimation of spatial pyramid matching (SPM) [49] SPP [18] net was proposed by He et al. to speed up R-CNN as well as to learn more discriminating features. The spatial Pyramid Pooling (SPP) layer is responsible for extracting fixed-length feature vectors. To address the limitations of SPP net, Fast R-CNN [41] was proposed by Girshick et al. in which features were extracted using the ROI pooling layer. Although the performance of fast R-CNN had improved, these traditional methods were used to generate proposals that are selective search [40]. This issue was addressed in Faster R-CNN [11] in which region proposal network (RPN) was introduced and it learned the features using the supervised learning method. Limitations of faster R-CNN were addressed in a region-based fully convolutional network (R-FCN) suggested by Dai et al. [15] in which computational cost is shared in the classification step. In a deep convolutional neural network, deep layers represent semantically strong features while shallow layers provide spatially strong features. This property is exploited by Lin et al. [24] and Feature

Table 1 Literature survey on vehicle detection and classification

Reference	Image Type	Dataset	Model	Accuracy (%)
[22]	Visible Images	LSVRC-2012	GMM+AlexNet6+PCA GMM+AlexNet6+LDA GMM+AlexNet7+PCA GMM+AlexNet7+LDA GMM+SIFT+PCA GMM+SIFT+LDA	96.75 97.80 91.40 96.30 96.25 96.45
[62]	Visible normal Images Visible dark images Visible transformed images Visible Late fusion images Visible images	Images were taken from a static camera	AlexNet-FC6 and 7+ SVM AlexNet-FC6+SVM AlexNet-FC6+SVM AlexNet-FC6+SVM Little YOLO-SPP Little YOLO-SPP Little YOLO-SPP MME-YOLO Extracted features contrast, homogeneity, entropy, and energy	97.35 74.68 79.32 85.41 77.44 52.95 47.41 91.63 92.7
[30]	Visible images	PASCAL VOC 2007, 2012 MS COCO 2014 PASCAL VOC+MS COCO Private Roadside dataset Images were taken by a visual camera		65.8
[64]	Visible images	Images were taken by FLIR ONE		70.5
[44]	Visible images	ImageNet VID Dataset	Feature-fused SSD detector and Tracking-guided detections optimizing (TDO)	93.63
[59]	Visible Images	Highway videos	Enhanced CNN+SVM	92.97
[48]	Visible Images	Custom dataset	SSD	92.97
[9]	Visible Images		MobileNet-SSD	79.3
[8]	Visible Images	CARPK and PUCPR+	ShuffleNet	62.89
[37]	Visible Images	MIO-TCO	Ensemble of global networks and Local expert networks	97.29
[42]	Visible Images	MIO-TCO	Ensemble of CNN with GANs	93.2

Pyramid Network was proposed in which semantically strong features were combined with spatially strong features so that objects at different scales could be detected.

One stage detection algorithm look at all positions on the image as probable objects. The first real-time one-stage detector was given by Redmon et al. [21] named You Only Look Once (YOLO) in which object detection was considered as a regression problem. A single shot detector (SSD) was proposed by Liu et al. [28] in 2016 to overcome the limitations of YOLO. In this, more than one feature map was utilized to predict an object, where each feature map was responsible for detecting an object of a specified scale according to its receptive fields. Hence, the network became complex to detect huge objects. Although SSD and Faster R-CNN are comparable in terms of accuracy, yet SSD is preferred for real-time detection due to its high inference speed. An amended version of YOLO that is YOLOv2 [36] is proposed by Redmon et al. in which anchors were generated by k-means clustering. Compared to YOLO, it had better accuracy as it was able to run efficiently on multiple scales and used a batch normalization layer.

CNN's are extremely efficient in visual understanding by learning semantic information of an image. VGG16 [15, 40], ResNet [15, 23], ResNeXt [63], and Hourglass [17] are backbone architectures trained on large datasets.

3 Proposed methodology

To train a deep learning model, 'N' annotated images $\{x_1, x_2, \dots, x_N\}$ are given and for i^{th} image x_i , there are M_i objects belonging to C categories:

$$y_i = \left\{ (c_1^i, b_1^i), (c_2^i, b_2^i), \dots, (c_{M_i}^i, b_{M_i}^i) \right\} \quad (1)$$

Where $c_j^i (c_j^i \in C)$ and b_j^i signifies categorical and spatial labels of the j^{th} object in x_i respectively. For x_i , the prediction y_{pred}^i shares the same format as y_i :

$$y_{pred}^i = \left\{ (c_{pred1}^i, b_{pred1}^i), (c_{pred2}^i, b_{pred2}^i), \dots \right\} \quad (2)$$

Over $C + 1$ categories, a multi-class classification model is trained, where C refers to target categories plus one background category.

3.1 Data collection

Training the vehicle detection model with a standard dataset like PASCAL VOC 2007, 2012 and MS COCO 2014 is not an ideal choice [30]. These datasets do not contain all types of vehicle categories. PASCAL VOC 2007 and 2012 consists of only two class labels that is car and bus. While MS COCO 2014 comprises 3 classes (car, bus, and truck). Therefore, to perform experiments four different datasets: FLIR thermal dataset, FLIR RGB dataset, MB7500, and KITTI dataset have been collected. FLIR dataset was released in 2018 which consists of approximately 14 K thermal as well as RGB images and 10 K videos [46]. The dataset comprises 60% daytime and 40% night time images From November to May when the weather is clear to overcast. The MB7500 dataset contains around 7500 photos obtained with a Phantom 4 drone and a high-definition camera in windy conditions. KITTI datasets are captured by driving around the mid-size city of Kalsruhe, in rural areas, and on highways.

3.2 Data annotation

Data annotation is the process of categorization and labelling of data. In this work, all the images are annotated using the labelling tool into six categories (cycle, two-wheeler, light vehicle, heavy vehicle, bus, and truck.)

3.3 Data augmentation

Data augmentation is also used to increase the data diversity for training models. In this work, three transformations have been applied to balance the dataset, which are horizontal flip, rotation, and Gaussian noise. Figure 1 shows an original image of a bus as well as augmented images.

3.4 Detection of vehicles using the ensemble method

In this paper, two DL models (Faster R-CNN and SSD) have been investigated. In some scenes, both the models are showing duplicate detections means the same object is detected in two categories at the same time, which leads to incorrect results [57]. Figure 2 shows duplicate detections made by Faster R-CNN in which a total of five objects are present. Out of five objects, one object is detected by the model as a heavy vehicle as well as a light vehicle; while other objects are not detected at all. Base models predictions have been improved by eliminating duplicate detections and further an ensemble using a majority voting classifier has been implemented.

3.4.1 Removal of duplicate detections in faster R-CNN and SSD

To remove the duplicate predictions of the Faster R-CNN and SSD model, obtained predictions are compared to each another based upon their bounding box coordinates. If the difference between two coordinates is less than the 25 (threshold = 25 taken using hit and trial method) then prediction with less confidence score is discarded.

3.4.2 Ensemble of Faster R-CNN and SSD

The technique of training numerous machine/deep learning models and integrating their outputs is known as ensemble learning. It's typically utilized to boost prediction performance, function approximation, and classification model accuracy. An ensemble of two deep learning models (Faster R-CNN and SSD) has been implemented in the proposed study. Faster R-CNN and SSD are explained as follows:



Fig. 1 Augmented Images (a) original (b) flipping (horizontal) (c) rotation (d) gaussian noise



Fig. 2 Prediction with Duplicate Detections by Faster R-CNN model

Faster R-CNN Faster R-CNN is a two-stage detector consisting of a total of three parts. (a) Convolution layers (b) Region Proposal Network (RPN) (c) Classes and bounding box prediction. The architecture of Faster R-CNN is shown in Fig. 3. CNN layers are used to extract features from the images. RPN predicts the possibility of the presence or absence of an object. It works on the last feature maps of the CNN layers and predicts the bounding box around the possible objects. The last part of the model is used to predict class labels and final bounding boxes.

3.5 Single shot detector (SSD)

SSD is a single-stage detector that consists of VGG-16 as a base without classification layer followed by multi-box convolutional layers. Figure 4 shows the architecture of the SSD model. The base model (VGG-16) is used to extract features. After the VGG-16 base model, various convolutional layers with decreasing size are placed to detect objects at varying scale and aspect ratios and to predict confidence scores. Initial layers are responsible for detecting small objects and deep layers detect larger objects. Hard negative mining is also utilized to avoid a large number of negative proposals.

Faster R-CNN and SSD model predictions are saved in Pfaster and Pssd, respectively. Predictions made by both models are compared based on bounding box coordinates to identify unique or duplicate detections. If coordinate values of two predictions have a difference of less than or equal to the threshold that means both the models have predicted a

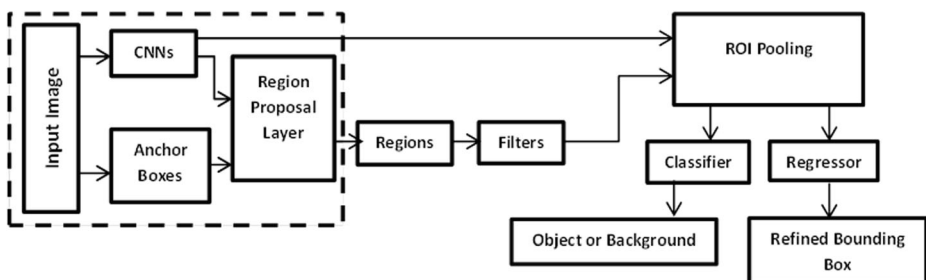


Fig. 3 Architecture of Faster R-CNN

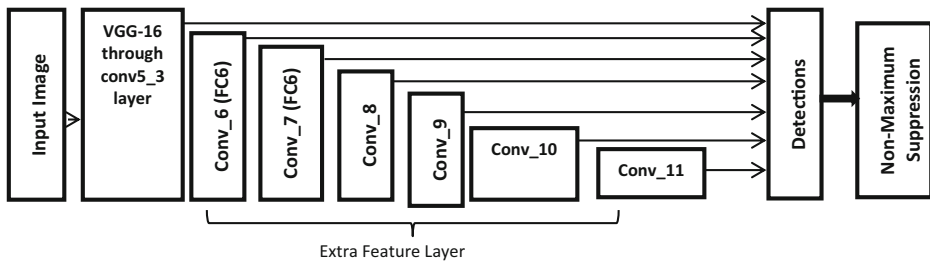


Fig. 4 Architecture of Single Shot Detector

specific vehicle. Then confidence score of P_{faster} is compared to the confidence score of P_{ssd} . If the confidence score of P_{faster} is greater than P_{ssd} then the prediction made by P_{faster} is saved to the final prediction and P_{ssd} prediction is discarded otherwise vice versa. A flow chart of the proposed technique is given in Fig. 5.

3.6 Traffic density estimation

Density is calculated in terms of the number of units present on-road rather than the number of vehicles. As each vehicle is having different size and shape. They occupy different areas as well as take different times to cross the intersection. In this paper, vehicles are categorized into six classes that are cycle, two-wheeler, light vehicles (small cars), heavy vehicles (big cars, vans, etc.), bus, and truck. Table 2 shows the number of units assigned to each type of vehicle.

Predictions made of each model are summed up by multiplying each vehicle with its corresponding units

$$\text{density} = \sum \text{predicted_category}_i * n_units \quad (3)$$

Where $\text{predicted_category}_i$ represents vehicle category predicted by the model and n_units represents the number of units associated with that category.

Algorithm 1 Proposed Methodology

INPUT: an image

OUTPUT: Predicted classes with bounding boxes and their confidence score

[P_{Ensemble} : bbox , class, conf]

Initialize the threshold value. And provide input images to model.

Begin

Obtain the results of base models. Suppose P_{faster} and P_{ssd} are the predictions obtained from faster R-CNN and SSD models which return bounding boxes coordinates, class of vehicle, and confidence score.

[P_{faster} : bbox, class, conf]

[P_{ssd} : bbox, class, conf]

If $P_{\text{faster}}.\text{bbox} - P_{\text{ssd}}.\text{bbox} \leq \text{threshold}$

 If $P_{\text{faster}}.\text{conf1} > P_{\text{ssd}}.\text{conf2}$

$P_{\text{Ensemble}}.\text{bbox} = P_{\text{faster}}.\text{bbox}$

$P_{\text{Ensemble}}.\text{class} = P_{\text{faster}}.\text{class}$

 Else

$P_{\text{Ensemble}}.\text{bbox} = P_{\text{ssd}}.\text{bbox}$

$P_{\text{Ensemble}}.\text{class} = P_{\text{ssd}}.\text{class}$

 End

End

End

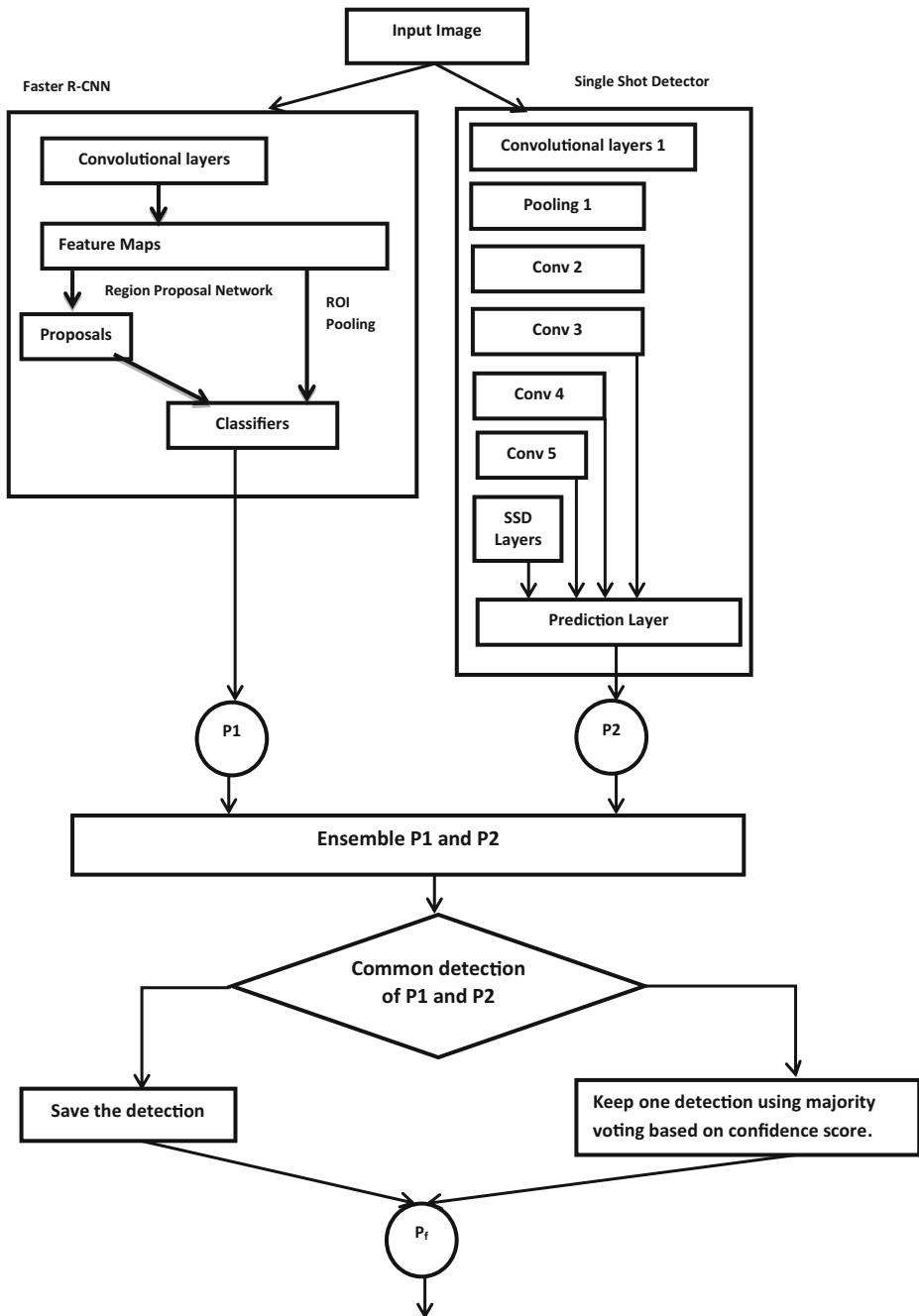


Fig. 5 Working of proposed DL ensemble

Table 2 Number of units assigned to different types of vehicles

Vehicle Type	Number of Units
Cycle	1
Two-Wheeler	2
Light Vehicle	4
Heavy Vehicle	6
Bus/Truck	10

4 Results and discussions

Implementation of all three models has been done in Python using TensorFlow API. The experimental platform configuration of the machine is given in Table 3. The input size of the model is 300 X 300 with a batch size of 24. The learning rate and IoU threshold are set to 0.0002 and 0.6 respectively. Table 4 shows the difference between thermal images and RGB images of the same scene taken from the FLIR dataset. As thermal images generate the images by sensing heat reflected by a body, the image quality is not affected by lighting and weather conditions. Thus, the visibility of objects is better in thermal images as compared to RGB.

4.1 Results of vehicle detection

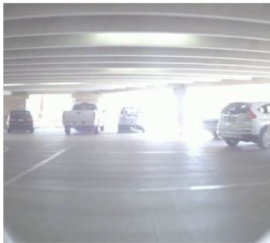
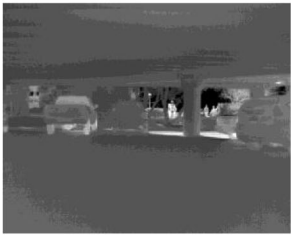
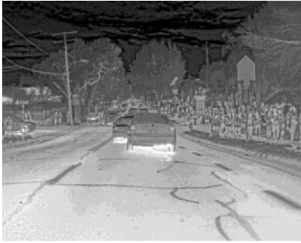
To compare the SSD, Faster R-CNN, and proposed models, precision vs. recall curve as well as (Mean average precision) mAP matrices have been used. The precision-recall curve (PR Curve) is useful in reporting information retrieval results. A good PR curve has a greater area under the curve (AUC). The mAP is useful for measuring the performance of object detection tasks. It compares the ground truth bounding box to the detected box. Figures 6, 7, 8 and 9 shows the graphs of the precision-recall curve of the FLIR thermal dataset, FLIR RGB, MB7500, and KITTI dataset respectively. All PR curves show that the proposed ensemble has better performance as compared to SSD and Faster R-CNN models. Table 5 shows the computed mAP values on the different datasets for SSD, Faster R-CNN, and proposed ensemble. The highest mAP obtained is 94% by proposed Ensemble on FLIR thermal dataset which is 34% better than SSD and 6% from Faster R-CNN model. Overall, the proposed ensemble achieves better and more promising results as compared to base estimators. Experimental results also depict that detection with thermal images is better than visible images. A comparison of different models on a collected dataset based on mAP is shown in Fig. 10.

Table 6 shows the comparison of the proposed work with existing studies. Till now, for vehicle detection, most of the work has been done using visible images. In this study, an ensemble of deep neural networks has been proposed and evaluated on thermal as well as visible images. The proposed model shows promising results on both types of images.

Table 3 Experimental Platform Configuration

Computing Machine	Configuration
Operating System	Windows 10
GPU	NVIDIA GEFORCE GTX (4GB)
RAM	8 GB
Processor	Intel Core i5
GPU acceleration library	CUDA, CUDNN

Table 4 RGB and Thermal Images at different times of a day

Time	RGB Images	Thermal Images
Day Time		
Excessive Sun Light		
Morning Time		
Night Time		
Evening Time		

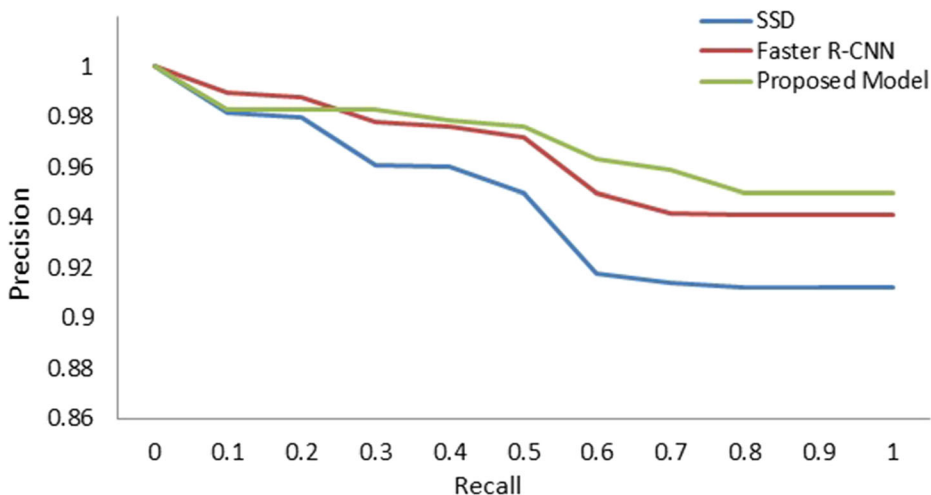


Fig. 6 Precision vs. Recall Curve of SSD, Faster R-CNN and Proposed model on FLIR Thermal dataset

4.2 Results of density estimation

Figures 11, 12, 13, 14, 15, 16, 17, 18 shows some of the experimental results obtained using SSD, Faster R-CNN, and proposed model on the collected datasets and analysis is made based on density estimated by each model. Computed density can be utilized for the designing of traffic light controllers.

Figure 11 consists of two heavy vehicles and two light vehicles. Figure 11a depicts the results of the SSD model in which the same vehicle is predicted as a heavy vehicle as well as a light vehicle means duplicate detection and SSD fails to detect other vehicles. While Fig. 11b and 11c show the results of Faster R-CNN and the proposed method respectively in which two

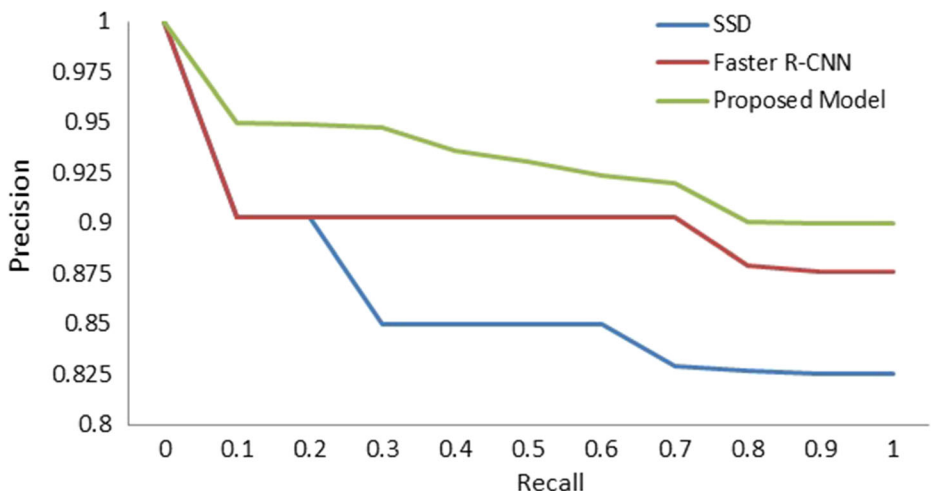


Fig. 7 Precision vs. Recall Curve of SSD, Faster R-CNN and Proposed model on FLIR RGB dataset

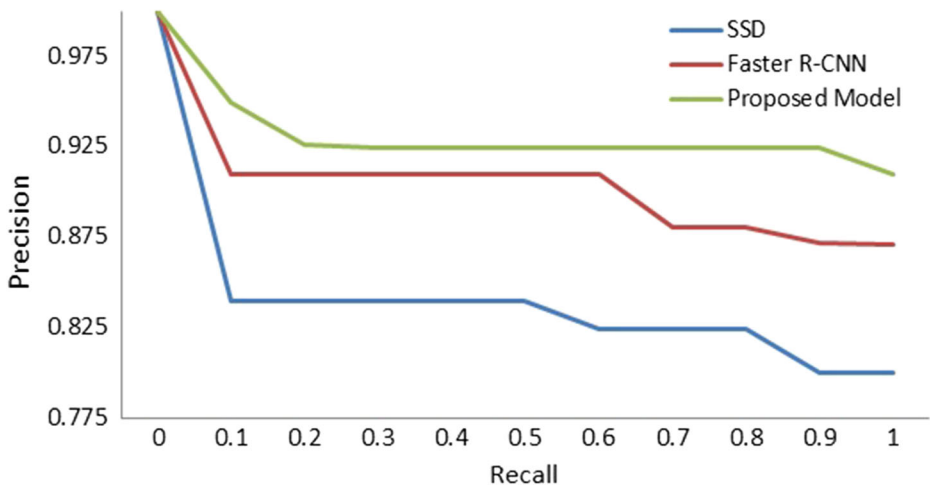


Fig. 8 Precision vs. Recall Curve of SSD, Faster R-CNN and Proposed model on MB7500 dataset

heavy vehicles are detected accurately and models fail to detect light vehicles. Hence, the actual density in the image is 20 (2 heavy vehicles = 12 units and 2 light vehicles = 8 units). Vehicle density computed by SSD models is 10 units, Faster R-CNN and proposed model computes 12 units respectively. SSD models also predict one object as a light vehicle as well as a heavy vehicle. Figure 12 contains one truck, two light vehicles, and two heavy vehicles and total estimated vehicle density is 30 units. Figure 12a depicts the results of the SSD model in which one truck and one light vehicle are detected and computed vehicle density is 14 units. Figure 12b shows the results of Faster R-CNN in which one truck and one heavy vehicle are detected and vehicle density is 16 units. Figure 12c contains the predictions made by proposing method. Vehicles are predicted as one truck, one light vehicle, and one heavy

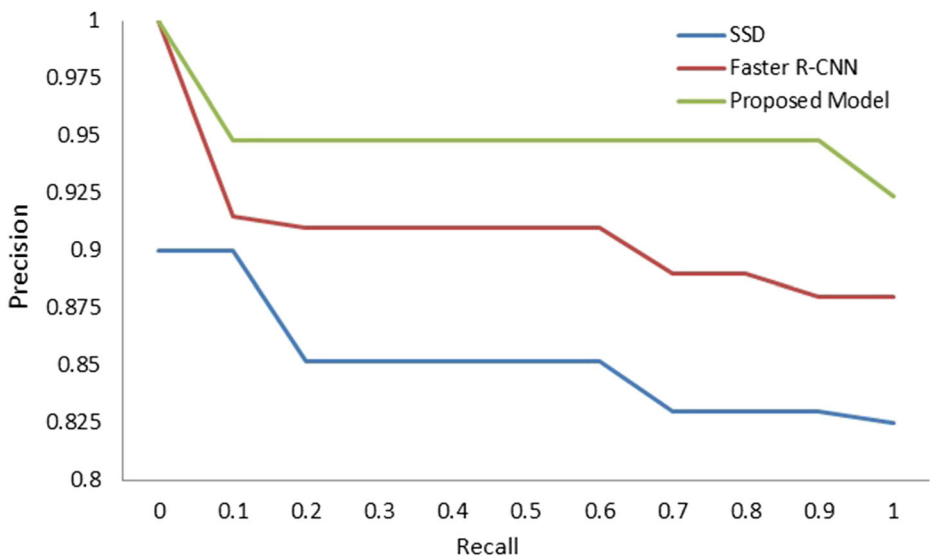


Fig. 9 Precision vs. Recall Curve of SSD, Faster R-CNN and Proposed model on KITTI dataset

Table 5 Comparative analysis of SSD, Faster R-CNN and Proposed ensemble based upon mAP on FLIR, KITTI, and MB7500

Model	Dataset	mAP	CY	TW	LV	HV	TR	BU
SSD	FLIR RGB	0.58	0.65	0.54	0.64	0.55	0.58	0.5
	FLIR Thermal	0.65	0.7	0.64	0.61	0.58	0.81	0.55
	KITTI	0.60	0.55	0.51	0.51	0.59	0.46	0.93
	MB7500	0.52	0.52	0.47	0.59	0.41	0.57	0.56
Faster R- CNN	FLIR RGB	0.87	0.94	0.87	0.8	0.94	1	0.67
	FLIR Thermal	0.88	0.96	0.91	0.86	0.87	0.86	0.8
	KITTI	0.81	0.75	0.75	0.84	0.89	0.77	0.85
	MB7500	0.68	0.68	0.6	0.59	0.94	0.6	0.67
Proposed Ensemble	FLIR RGB	0.92	0.94	0.93	0.91	0.91	1	0.83
	FLIR Thermal	0.94	0.95	0.96	0.89	0.97	1	0.9
	KITTI	0.92	0.85	0.87	0.96	0.92	0.97	0.98
	MB7500	0.86	0.89	0.8	0.96	0.94	0.78	0.78

vehicle, and computed vehicle density is 20 units which are the closest value to true value among all models.

Figure 13 shows that there are two vehicles present in the image that is light vehicle and a truck. Thus, the total vehicle density is 14 units. Figure 13a contains detections made by the SSD model, where it detects both the vehicles and label them correctly as light vehicle and truck. Thus, the computed vehicle density is 14 units. While Faster R-CNN predicts only light vehicles and misses a truck. So, computer density is 4 units only. Predictions made by the

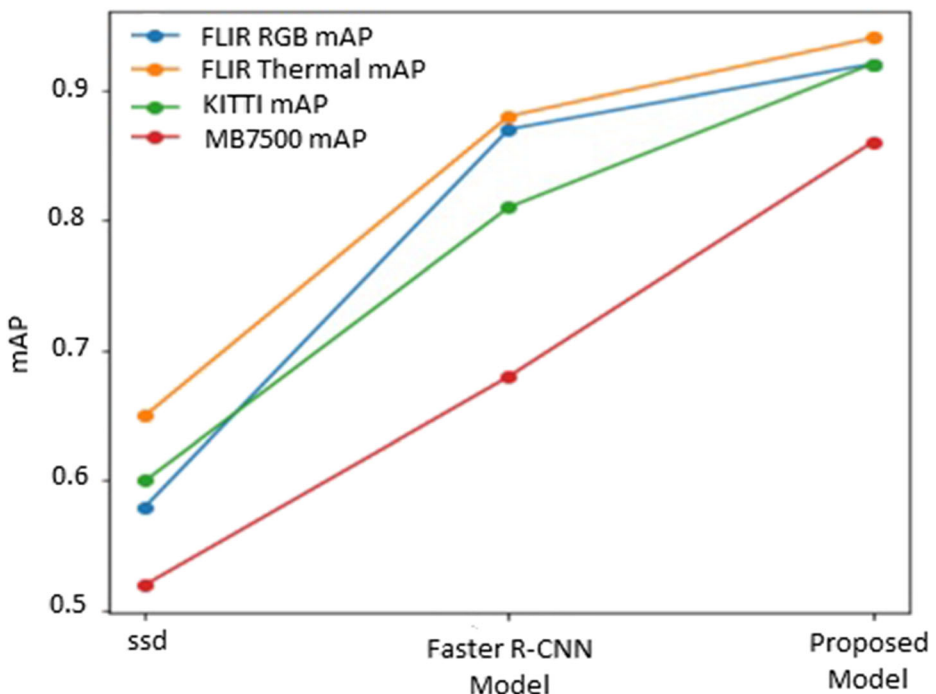
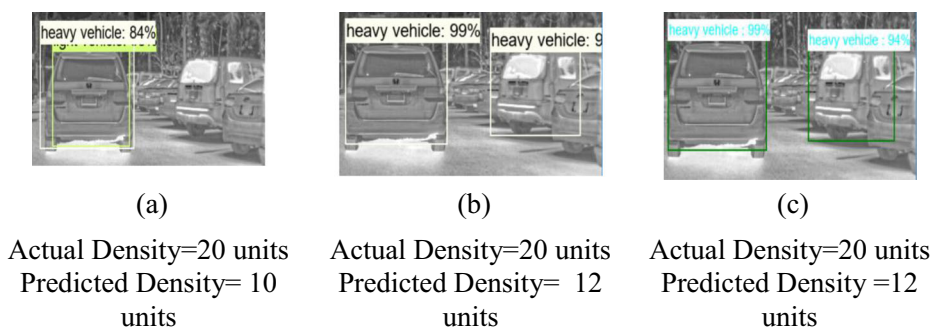
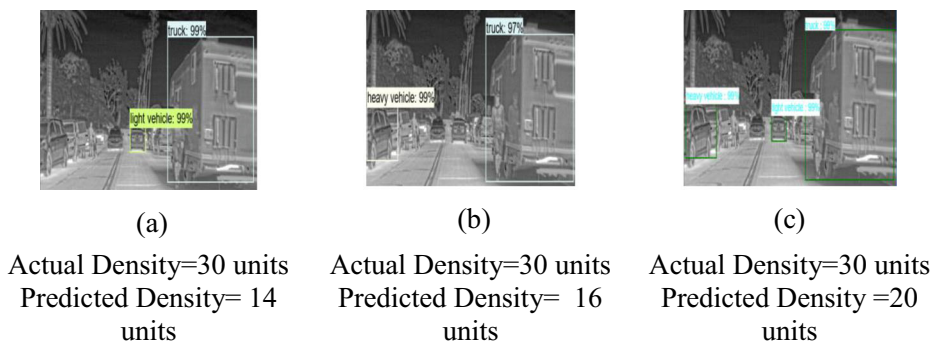
**Fig. 10** mAP of SSD, Faster R-CNN and Proposed Ensemble on FLIR RGB, Thermal, KITTI, and MB7500 dataset

Table 6 Comparison of the proposed model with existing methods

Technique/Reference	Image Type	Accuracy
[44]	Visible Images	92.7
[44]	Thermal Images	65.8
[42]	Visible Images	93.2
[8]	Visible Images	62.89
[48]	Visible Images	93.63
[30]	Visible Images	77.44
Proposed Model	Visible Images	92
Proposed Model	Thermal Images	94

proposed method are correct and the estimated density is 14 units. Figure 14 consists of two light vehicles and one heavy vehicle. Therefore, the actual vehicle density is 14 units. Figure 14a depicts results of SSD models showing only one vehicle as a light vehicle and vehicle density as 4 units. While Fig. 14b and 14c show the results of Faster R-CNN and the proposed method respectively in which all vehicles are predicted accurately, and computed vehicle density is 14 units.

Figure 15 has five vehicles one truck, one bus, two light vehicles, and one 2-wheeler. Thus, the actual density is 30 units. Results of the SSD model are shown in Fig. 15a and SSD predicts one bus correctly. A truck is detected as a light vehicle. Thus, the total predicted density is 14. Faster R-CNN predicted a 2-wheeler as one cycle and computed density is only 1

**Fig. 11** Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t FLIR thermal dataset**Fig. 12** Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t FLIR thermal dataset



(a)

Actual Density=14 units
Predicted Density=14
units



(b)

Actual Density=14 units
Predicted Density= 4
units



(c)

Actual Density=14 units
Predicted Density =14
units

Fig. 13 Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t FLIR RGB dataset

unit. Finally, the proposed model predicts three vehicles and predicted is 15 units. Figure 19 has four vehicles, that is one light vehicle, one bus, and two 2-wheelers. Thus, the actual density is 18 units. Results of the SSD model are shown in Fig. 16a and SSD predicts only one light vehicle and the estimated vehicle density is 4 units. Where predictions made by Faster R-CNN and the proposed method are given in Figs. 16b and 16c respectively. These models



(a)

Actual Density=14 units
Predicted Density= 4
units



(b)

Actual Density=14 units
Predicted Density= 14
units



(c)

Actual Density=14 units
Predicted Density =14
units

Fig. 14 Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t FLIR RGB dataset



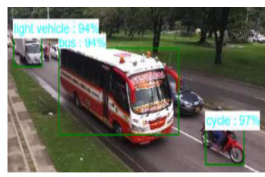
(a)

Actual Density=30 units
Predicted Density= 14
units



(b)

Actual Density=30 units
Predicted Density= 1
units



(c)

Actual Density=20 units
Predicted Density =15
units

Fig. 15 Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t MB7500 dataset

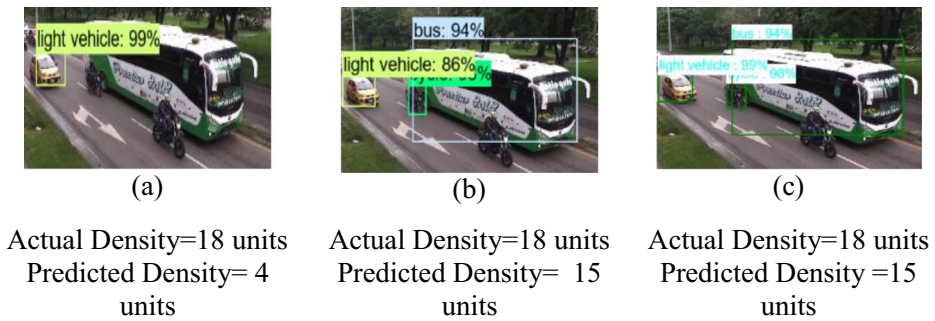


Fig. 16 Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t MB7500 dataset

detect one light vehicle, one bus, and one 2-wheeler is predicted as a cycle. Therefore, the estimated vehicle density is 15 units for both models.

Figure 17 consists of one cycle and one heavy vehicle total 7 units of density. SSD has not predicted any vehicle. Thus, the estimated density is 0 units. While Figs. 17b and 17c show the results of Faster R-CNN and proposed model respectively in which cycle as well as heavy vehicle is predicted correctly, and computed density is 7 units. Figure 18 contains one heavy vehicle and one light vehicle. Thus, the actual vehicle density is 10 units. Figure 18a shows the results of the SSD model in which only one heavy vehicle is detected, and the computed

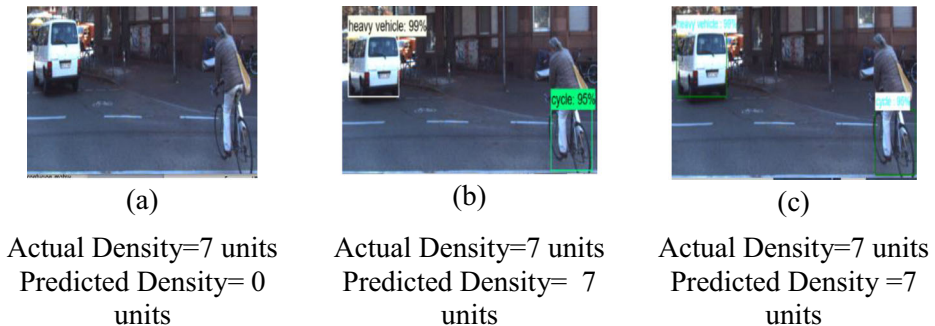


Fig. 17 Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t KITTI dataset

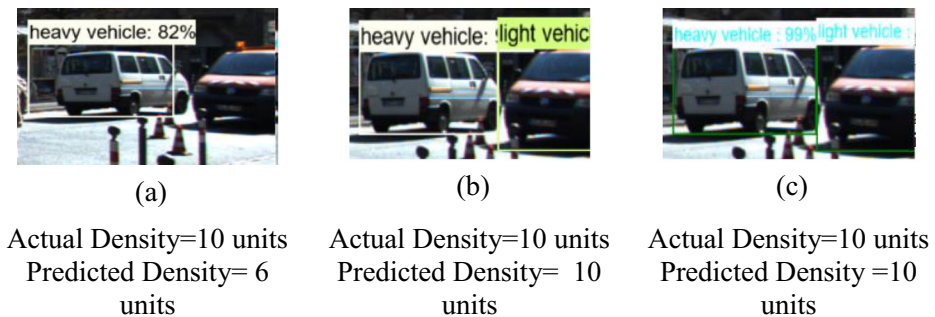


Fig. 18 Actual and predicted density of (a) SSD model, (b) Faster R-CNN, and (c) Proposed Ensemble w.r.t KITTI dataset

Table 7 Comparison of inference speed of Faster R-CNN, SSD, and Proposed Method

Model	Inference Speed
Faster R-CNN	15.67 Sec
SSD	10.15 Sec
Ensemble	22.37 Sec

vehicle density is 6 units. While Fig. 18b and 18c show the results of Faster R-CNN and proposed model respectively in which both the vehicle detected and computed density is the same as actual density.

Other than the detection accuracy, the effectiveness of the detection model can also be estimated using inference speed. Table 7 shows the inference speed of different models SSD, Faster R-CNN, and the proposed method.

Considering the detection results and inference speed together, it is concluded that the proposed model is providing better results in terms of detection and density calculation; although the time is taken to process the image is high as compared to other models.

5 Conclusion

An ensemble-based deep learning architecture has been presented and analysed in this paper. The present scheme is a collaboration Faster R-CNN and SSD for object detection and majority voting concept has been used which improves overall predictions. Although two deep learning models have been used which increases the time complexity of the model, yet it increases overall accuracy which is acceptable. Besides the detection and classification task, traffic density estimation is also done which gives the occupancy rate of the road at a particular time. The comparison of our results with base models shows that the proposed model is capable of offering better results. Also, the proposed model efficiently resolves the problem of duplicate predictions based on confidence scores. Thus, the proposed model can be used in intelligent transport applications for the detection of vehicles and optimization of signal controllers based upon density. One of the major limitations of the proposed work is its running time complexity optimization time and computational complexity optimization.

Declarations

Conflict of interest There is no conflict of interest.

References

1. AlZu'bi S, Jararweh Y (2020) Data fusion in autonomous vehicles research, literature tracing from imaginary idea to smart surrounding community. In: Fifth International Conference on Fog and Mobile Edge Computing (FMEC), pp 306–311. <https://doi.org/10.1109/FMEC49853.2020.9144916>
2. AlZu'bi S, Al-Qatawneh S, Alsmirat M (2018) Transferable HMM trained matrices for accelerating statistical segmentation time. In: Fifth International Conference on Social Networks Analysis, Management and Security (SNAMS), pp 172–176. <https://doi.org/10.1109/SNAMS.2018.8554487>
3. AlZu'bi S, Hawashin B, Mujahed M, Jararweh Y, Gupta BB (2019) An efficient employment of internet of multimedia things in smart and future agriculture. *Multimed Tools Appl* 78(20):29581–29605. <https://doi.org/10.1007/s11042-019-7367-0>

4. AlZu'bi S, Shehab M, Al-Ayyoub M, Jararweh Y, Gupta B (2020) Parallel implementation for 3D medical volume fuzzy segmentation. *Pattern Recogn Lett* 130:312–318. <https://doi.org/10.1016/j.patrec.2018.07.026>
5. Al-Zu'bi S, Hawashin B, Mughaid A, Baker T (2020) Efficient 3D medical image segmentation algorithm over a secured multimedia network. *Multimed Tools Appl* 80(11):16887–16905. <https://doi.org/10.1007/s11042-020-09160-6>
6. Aql D, Al-Zubi S, Mughaid A, Jararweh Y (2021) Extreme learning machine for plant diseases classification: a sustainable approach for smart agriculture. *Clust Comput* 25(3):2007–2020. <https://doi.org/10.1007/s10586-021-03397-y>
7. Arrospe J, Salgado L (2013) Log-Gabor filters for image-based vehicle verification. *IEEE Trans Image Process* 22(6):2286–2295. <https://doi.org/10.1109/tip.2013.2249080>
8. Azimi SM (2018) ShuffleDet: real-time vehicle detection network in on-board embedded UAV imagery. In: European Conference on Computer Vision ECCV. Computer vision – ECCV 2018 workshops Lect Notes Comput Sci 2019:88–99. https://doi.org/10.1007/978-3-030-11012-3_7
9. Biswas D, Su H, Wang C, Stevanovic A, Wang W (2019) An automatic traffic density estimation using Single Shot Detection (SSD) and MobileNet-SSD. *Phys Chem Earth, Parts* 110:176–184. <https://doi.org/10.1016/J.PCE.2018.12.001>
10. Bodla N, Singh B, Chellappa R, Davis LS (2017) Soft-nms – improving object detection with one line of code. In: Proceedings of the IEEE international conference on computer vision, pp 5561–5569
11. Cai Z, Vasconcelos N (2018) Cascade r-cnn: delving into high quality object detection. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 6154–6162
12. Chan YM, Huang SS, Fu LC, Hsiao PY, Lo MF (2012) Vehicle detection and tracking under various lighting. *IET Intell Transp Syst* 6:1–8
13. Chen Y, Li J, Xiao H, Jin X, Yan S, Feng J (2017) Dual path networks. *Adv Neural Inf Process Syst* 30
14. Chowdhury PN, Shivakumara P, Pal U, Lu T, Blumenstein M (2020) A new augmentation-based method for text detection in night and day license plate images. *Multimed Tools Appl* 79(43–44):33303–33330. <https://doi.org/10.1007/s11042-020-09681-0>
15. Dai J, Li Y, He K, Sun J (2016) R-fcn: object detection via region-based fully convolutional networks. *Adv Neural Inf Process Syst* 29
16. Dalal N, Triggs B (2005) Histograms of oriented gradients for human detection. In: 2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05). Ieee, vol 1, pp 886–893
17. Duan K, Bai S, Xie L, Qi H, Huang Q, Tian Q (2019) Centernet: Keypoint triplets for object detection. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 6569–6578
18. Fidler S, Mottaghi R, Yuille A, Urtasun R (2013) Bottom-up segmentation for top-down detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp 3294–3301
19. Gao Y, Lee HJ (2015) Moving car detection and model recognition based on deep learning. *Adv Sci Technol Lett* 90 57–61
20. Girshick R (2015) Fast r-cnn. In: Proceedings of the IEEE international conference on computer vision, pp 1440–1448
21. Girshick R, Donahue J, Darrell T, Malik J (2014) Rich feature hierarchies for accurate object detection and semantic segmentation. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 580–587
22. Harsha SS, Anne KR (2016) Gaussian mixture model and deep neural network based vehicle detection and classification. (IJACSA). *Int J Adv Comput Sci Appl* 7(9):17–25
23. He K, Zhang X, Ren S, Sun J (2014) Spatial pyramid pooling in deep convolutional networks for visual recognition. *IEEE Trans Pattern Anal Mach Intell* 37(9):1904–1916
24. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 770–778
25. He K, Zhang X, Ren S, Sun J (2016) Identity mappings in deep residual networks. In: European conference on computer vision. Springer, Cham, pp 630–645
26. Howard AG, Zhu M, Chen B, Kalenichenko D, Wang W, Weyand T, Andreetto M, Adam H (2017) Mobilenets: efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv: 1704.04861*
27. Huang G, Liu Z, Van Der Maaten L, Weinberger KQ (2017) Densely connected convolutional networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 4700–4708
28. Ioffe S, Szegedy C (2015) Batch normalization: accelerating deep network training by reducing internal covariate shift. In: International conference on machine learning. PMLR, pp 448–456
29. Jagannathan P, Rajkumar S, Frnda J, Divakarachari PB, Subramani P (2021) Moving vehicle detection and classification using Gaussian mixture model and ensemble deep learning technique. *Wirel Commun Mob Comput* 2021:1–15. <https://doi.org/10.1155/2021/5590894>

30. Jamiya SS, Esther Rani P (2021) LittleYOLO-SPP: A delicate real-time vehicle detection algorithm. *Optik* 225:165818. <https://doi.org/10.1016/j.jlleo.2020.165818>
31. John V, Mita S, Zheng L, Qi B (2015) Pedestrian detection in thermal images using adaptive fuzzy C-means clustering and convolutional neural networks. In: 2015 14th IAPR International Conference on Machine Vision Applications (MVA). IEEE, pp 246–249
32. Kembhavi A, Harwood D, Davis LS (Jun. 2011) Vehicle detection using partial least squares. *IEEE Trans Pattern Anal Mach Intell* 33(6):1250–1265. <https://doi.org/10.1109/TPAMI.2010.182>
33. Keserwani P, Dhankhar A, Saini R, Roy PP (2021) Quadbox: quadrilateral bounding box based scene text detection using vector regression. *IEEE Access* 9:36802–36818. <https://doi.org/10.1109/ACCESS.2021.3063030>
34. Kleban J, Xie X, Ma W-Y (2008) Spatial pyramid mining for logo detection in natural scenes. In: 2008 IEEE International Conference on Multimedia and Expo. IEEE, pp 1077–1080
35. Krizhevsky A, Sutskever I, Hinton GE (2012) Imagenet classification with deep convolutional neural networks. *Adv Neural Inf Process Syst* 25
36. Law H, Deng J (2018) Cornernet: detecting objects as paired keypoints. In: Proceedings of the European conference on computer vision (ECCV), pp 734–750
37. Lee JT, Chung Y (2017) Deep learning-based vehicle classification using an ensemble of local expert and global networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition workshops, pp 47–52. <https://doi.org/10.1109/CVPRW.2017.127>
38. Lienhart R, Maydt J (2002) An extended set of haar-like features for rapid object detection. In: Proceedings international conference on image processing. IEEE, vol 1, pp I–I
39. Lin T-Y, Dollár P, Girshick R, He K, Hariharan B, Belongie S (2017) Feature pyramid networks for object detection, in: CVPR
40. Lin T-Y, Goyal P, Girshick R, He K, Dollár P (2017) Focal loss for dense object detection, in: ICCV
41. Liu W, Anguelov D, Erhan D, Szegedy C, Reed S, Fu C-Y, Berg AC (2016) SSD: single shot multibox detector, in: ECCV
42. Liu W, Luo Z, Li S (2018) Improving deep ensemble vehicle classification by using selected adversarial samples. *Knowl-Based Syst* 160:167–175. <https://doi.org/10.1016/j.knosys.2018.06.035>
43. Lowe DG (1999) Object recognition from local scale-invariant features, in: ICCV
44. Nam Y, Nam Y-C (2018) Vehicle classification based on images from visible light and thermal cameras. *EURASIP J Image Vid Process* 2018. <https://doi.org/10.1186/s13640-018-0245-2>
45. Newell A, Yang K, Deng J (2016) Stacked hourglass networks for human pose estimation. In: European conference on computer vision. Springer, Cham, pp 483–499
46. Oliveira DC, Wehrmeister MA (2018) Using deep learning and low cost RGB and thermal cameras to detect pedestrians in aerial images captured by multirotor UAV. *Sensors (Basel)* 18(7):2244. <https://doi.org/10.3390/s18072244>
47. Ozkurt C, Camci F (Dec. 2009) Automatic traffic density estimation and vehicle classification for traffic surveillance systems using neural networks. *Math Comput Appl* 14(3):187–196. <https://doi.org/10.3390/mca14030187>
48. Ranjeeth Kumar C, Anuradha R (2020) Feature selection and classification methods for vehicle tracking and detection. *J Ambient Intell Humaniz Comput* 12(3):4269–4279. <https://doi.org/10.1007/s12652-020-01824-3>
49. Redmon J, Farhadi A (2017) Yolo9000: better, faster, stronger, in: CVPR
50. Redmon J, Divvala S, Girshick R, Farhadi A (2016) You only look once: unified, real-time object detection, in: CVPR
51. Ren S, He K, Girshick R, Sun J (2015) Faster r-cnn: towards real-time object detection with region proposal networks, in: NeurIPS
52. Rezaee H, Aghagholizadeh A, Seyedarabi MH, Zu'bi SA (2011) “Tracking and occlusion handling in multi-sensor networks by particle filter,” in IEEE GCC Conference and Exhibition (GCC), pp. 397–400, <https://doi.org/10.1109/IEEEGCC.2011.5752541>.
53. Simonyan K, Zisserman A (2014) Very deep convolutional networks for large-scale image recognition, in: arXiv preprint arXiv:1409.1556
54. Suryanto, Kim D-H, Kim H-K, Ko S-J (2011) Spatial color histogram based center voting method for subsequent object tracking and segmentation. *Image Vis Comput* 29(12):850–860. <https://doi.org/10.1016/j.imavis.2011.09.008>
55. Szegedy C, Liu W, Jia Y, Sermanet P, Reed S, Anguelov D, Erhan D, Vanhoucke V, Rabinovich A (2015) Going deeper with convolutions, in: CVPR
56. Uijlings JR, Van De Sande KE, Gevers T, Smeulders AW (2013) Selective search for object recognition. *IJCV* 104:154–171
57. Wu X, Sahoo D, Hoi SCH (2019) Recent advances in deep learning for object detection. Elsevier

58. Xie S, Girshick R, Dollár P, Tu Z, He K (2017) Aggregated residual transformations for deep neural networks, in: CVPR
59. Yang Y, Song H, Sun S, Zhang W, Chen Y, Rakal L, Fang Y (2021) A fast and effective video vehicle detection method leveraging feature fusion and proposal temporal link. *J Real-Time Image Proc* 18(4): 1261–1274. <https://doi.org/10.1007/s11554-021-01121-y>
60. Zhang B (2013) Reliable classification of vehicle types based on Cascade classifier ensembles. *IEEE Trans Intell Transp Syst* 14(1):322–332. <https://doi.org/10.1109/tits.2012.2213814>
61. Zhang S, Wen L, Bian X, Lei Z, Li SZ (2018) Single-shot refinement neural network for object detection, in: CVPR
62. Zhou Y, Nejati H, Do TT, Cheung NM, Cheah L (2016) Image-based vehicle analysis using deep neural network: a systematic study. *IEEE international conference on digital signal processing (DSP)*. IEEE, Beijing
63. Zhou X, Wang D, Krähenbühl P (2019) Objects as points, in: arXiv preprint arXiv:1904.07850
64. Zhu J, Li X, Jin P, Xu Q, Sun Z, Song X (2020) MME-YOLO: Multi-Sensor Multi-Level Enhanced YOLO for Robust Vehicle Detection in Traffic Surveillance. *Sensors* 21(1):27. <https://doi.org/10.3390/s21010027>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.